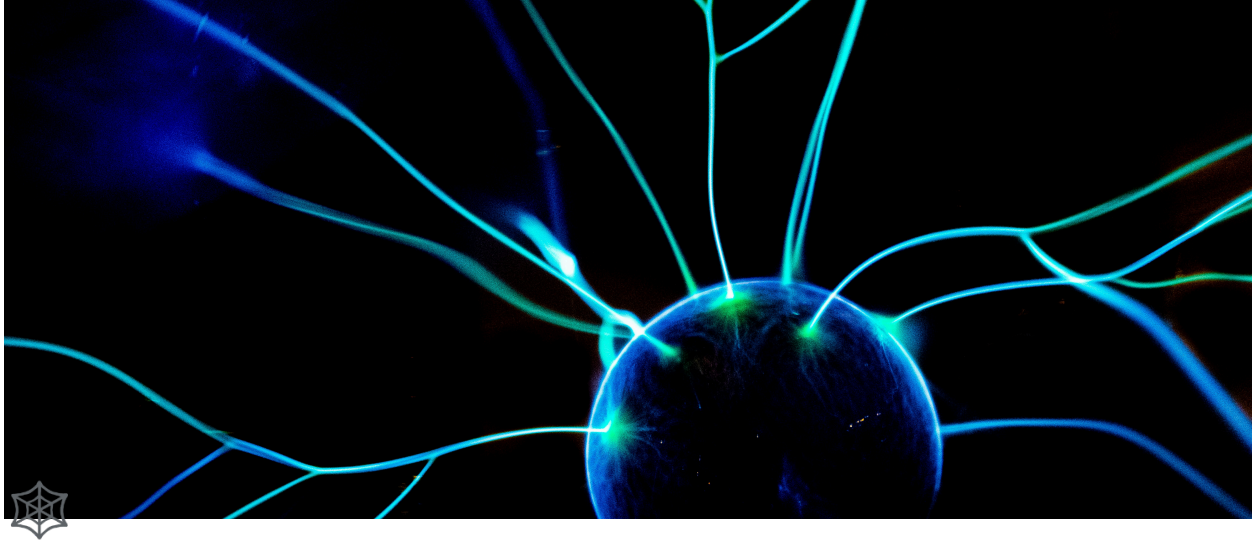




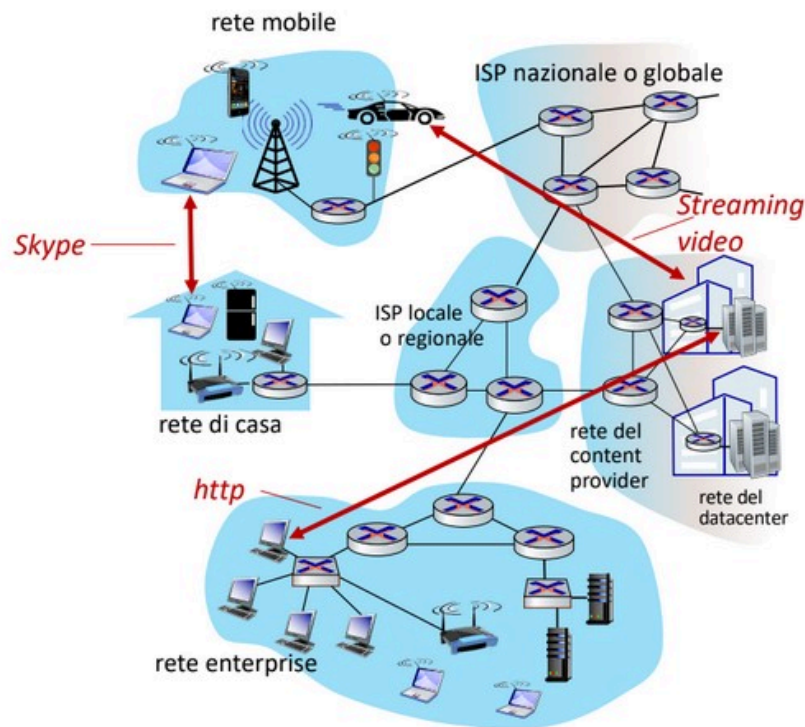
Computer networks (non completi)

What is a connection?

A connection is an open communication where there are parts that attend for messages and parts that send messages. In this type of connection the communication protocol is very important to avoid the risk to exchange incomprehensible messages.

What is switch, host, network, internet and Internet?

- Host: is a peripheral device connected to the network (pcs, smartphones, printers)
- Switch: is a device that can route information
- Network: a collection of Host/Switch
- internet: a network of network
- Internet: all the existent internet interconnected.



Communication Protocol

It is a set of rules that a device has to follow to communicate with other devices (send/receive messages).

In the communication protocol is also defined the information format.

Very simple example:

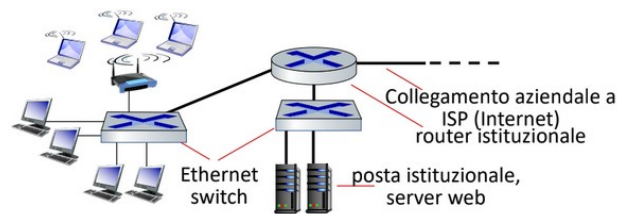
When we talk with others people we implicitly follow a communication protocol.

- A asks: "what time is it?" to B.
B responds: "it's half past three" to A.
A responds: "thanks" to B.
- A asks: "What time is it?" to B.
B doesn't respond to A (because A is a very suspicious man).
10 second passed..
A asks: "What time is it?" to B.
B doesn't respond to A (because A is a very suspicious man).
A asks: "What time is it?" to C.
C responds: "it's half past three" to A.

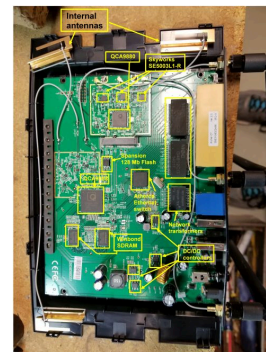
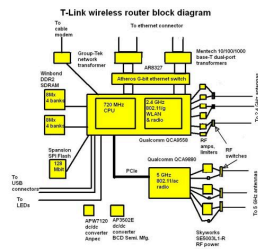
Internet structure

A network is a set of nodes (Router, Host, switch, server...) linked by links. in common language we call:

- Server : a device that delivers services.
- Router : Interconnection device that connect a network to another network (your home network to Lazio's ISP)
- Modem : is a device that "traslates" analogical input from ISP to digital output for hosts and the revers. (MOD: mudulator, DEM: demodulator).
- Host : a device that use a service.
- Switch : device that connect more devices to a local network (your phone, your pc to your home network) in the last decade wireless switches are very common due the large quantity of smartphones.



in the last decade switch, modem and router can be integrated on the same device.

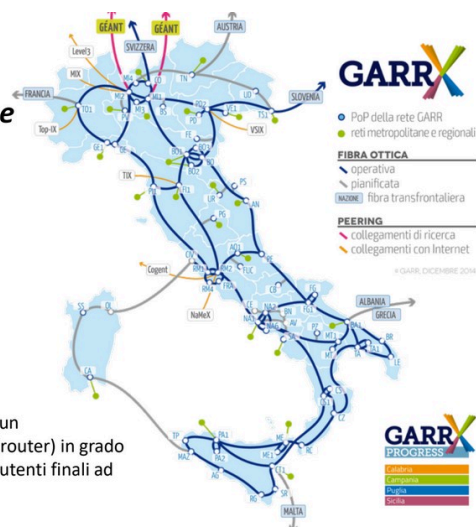


Links between networks can be built in copper or in optical fiber, but, they also can be wireless (electromagnetic waves).

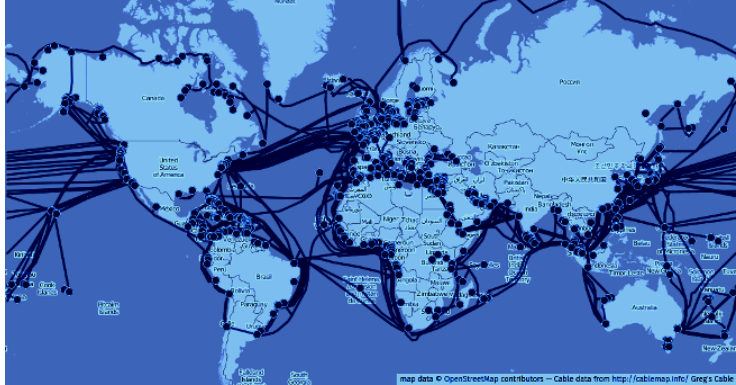
Local network (home network) is linked to a regional ISP (internet service provider) consisting of interconnected routers that form the “core” or “backbones”.

This regional ISP is connected to a national ISP and this is connected to other world’s internet, this is **Internet**. link between ISP is named IXP.

GARR
Gruppo per
l'Armonizzazione
delle Reti
della Ricerca



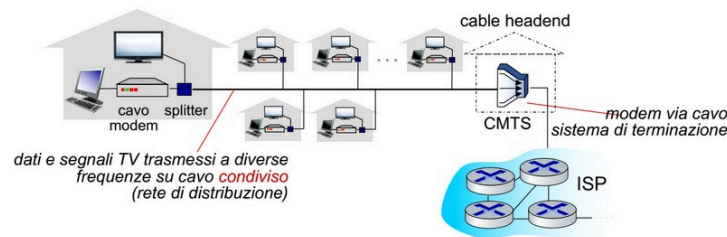
POP (Point of Presence) è un punto di accesso alla rete (router) in grado di instradare il traffico agli utenti finali ad esso connessi



Types of cable networks

- TELEVISION.

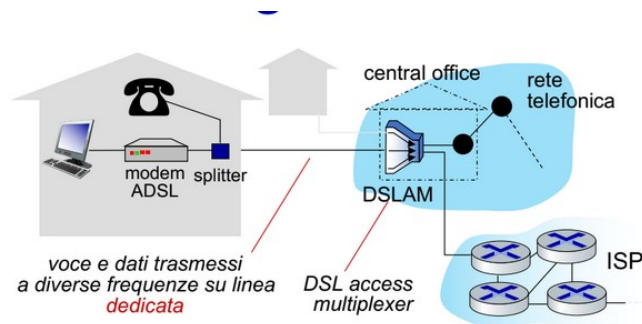
In the cable networks there is a CMTS (Cable, Modem, Termination, System) connected to a local ISP and connected to various houses/offices. In each home/office there is a modem that “translates” (modules/de-modules) signals and a splitter that abilities the use of the television or the use of internet connection.



In this case the information on the cable are in common for each home/office.

- DSL

The DSL (Digital Subscriber Line) is different, it is use for internet and works thanks to the telephone line (for this reason it is very slow). There is a DSLAM that is connect to a local ISP, to telephonic network and to various houses/offices. In each house/office there is an ADSL modem that modules/de-modules the signal from or to DSLAM and a splitter that abilities the use of the telephone or the use of internet connection.



In each house there is a modem, modem is connected to various switch, these switches can be connected to the hosts with various type of technologies:

- ethernet cable (fastest)
- WI-FI access point (slowest)

- Wide Area LAN

Is the common mobile internet connection, the mobile operators use big antennas to have big signal coperture.

Transmission

Hosts communicate with servers receiving and sending messages in data packet form. The messages is divided in data packet (the data packet size can vary between protocols)

- Hosts and servers exchange information with velocity of transmission = R (bit/sec)
- Transmission retard is the time to transmit a packet into the link (packet size/ R)
 - Types of transmission:
 - coaxial cable: is copper bidirectional cable, it very resistant to interferences.
 - optical fiber cable: it transport information in light form, for this reason is very important to create a good network of repeater to prevent the information deterioration. (fastest)
 - wireless : the signal propagates through the air, for this reason it is less resistant to interferences. Is not a bidirectional solution because if an antenna is sending a message it cannot receives a message from other antennas, the outgoing signal interferizes with the incoming signals.

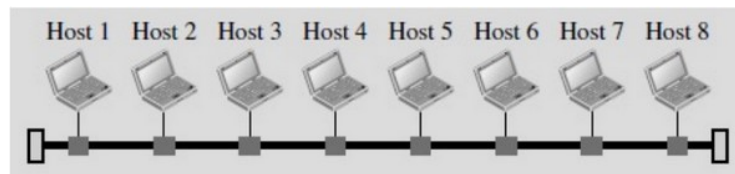
There exist a network classification scale:

Scale	Type	Example
Vicinity	PAN (Personal Area Network)	Bluetooth (e.g., headset)
Building	LAN (Local Area Network)	WiFi, Ethernet
City	MAN (Metropolitan Area Network)	Cable, DSL
Country	WAN (Wide Area Network)	Large ISP
Planet	The Internet (network of all networks)	The Internet!

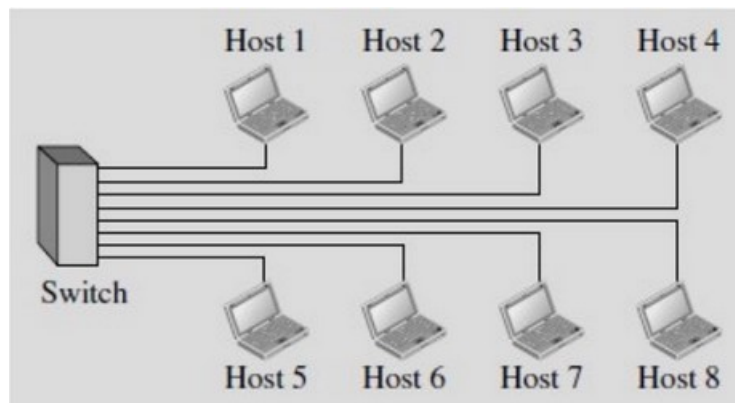
LAN and WAN

[Lan](#) is a local network, it can be “common cable” where all the devices are connected with the same cable, or “switching connection” where each device (host) is connected to a switch with own cable, it is more secure and faster because the traffic in cables is limited to only one device per cable.

COMMON CABLE:



SWITCHING CONNECTION:



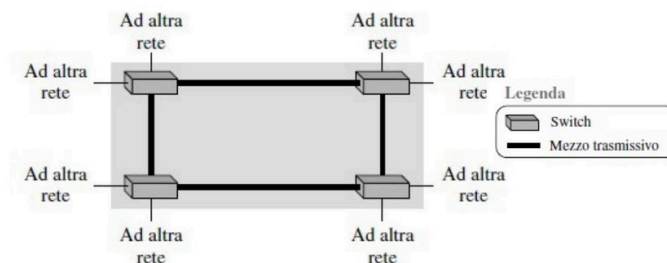
[Wan](#) is more large network, regional/statal network, this type of network is managed by big telecommunication operators. Wan can be “switching connection” or “point to point”, In point to point connection only two smaller

networks can be connected while in switching connection type, more networks can be connected using switches.

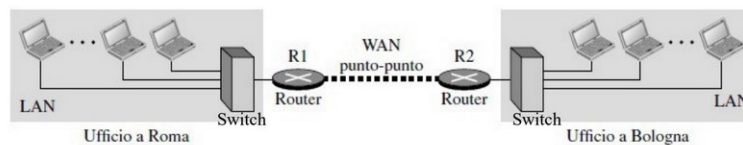
Esempio di WAN punto-punto



WAN a commutazione (switched)



Wan example:

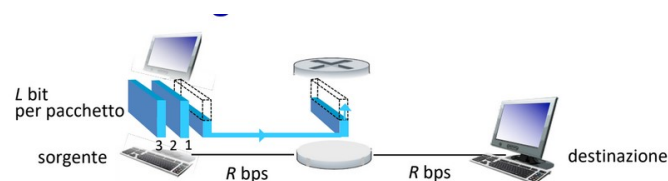


Network's kernel

Network's kernel is a network of interconnected router.

All the information on an internet pass through this backbones following these passages.

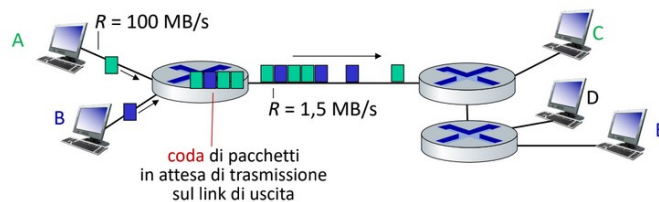
- packet commutation.
Hosts divided informations in packets and they are sent through this kernel from the first router (personal router) to the last (server's router).
- routing.
routing is the action of route the packets to the destination, this action follow various route algorithm.
- transimtion.
transimtion is the action to trafer a packet in the communication channel (not the action to trafer the information to the destinator)
 - It has a delay, transimtion delay and this is: $((\text{bit per packet "L"}) / \text{bit per second "R"})$.



- store and forward

routers work in this way: a packet has to be fully loaded on a router to be re-transmitted.

- A common problem due this policy is the “queuing”.

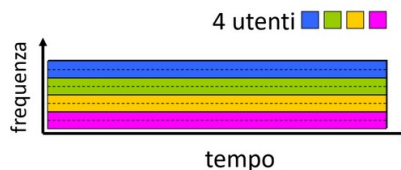


Packets can be re-transmitted to the link only if they are fully loaded in router's RAM due to this, if the packets are too much and the router's RAM is full, there is a “packet loss”.

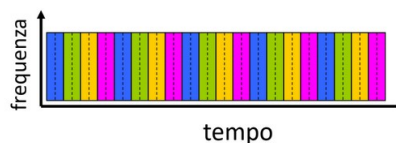
- A possible solution to solve this problem is a “circuit switching”, where each host has its own link to the router, this type of router is very expensive and links can be inutilized.

Another problem is that only a limited number of hosts can be connected to the same router.

- A more cheap solution is to divide the same link to different hosts in an efficient way.
 - FDM (frequency division multiplexing)
hosts divided by frequency:
 - benefit: multi-host operation.
 - malus: no large-band (no full speed connection).



- TDM (time divisione multiplexing)
time divided in SLOTS, each host has a slot at disposition to sends/recieves packets.
 - benefit: large-band (full speed connection).
 - malus: more turnaround time.
single-host operation.



Packets-loss is inevitable it can only reduced with this tecnics.

(In real life all these three tecnics are used simultanesly)

Probability to have a packets-loss:

- k is the probability that all the user communicate simultanesly.
- p is the probability that an host communicates
- n is the number of hosts

$$\Pr(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Example: In a system with 35 hosts we have packets-loss if 10 hosts are sending packets at the same moment, what is the probability to have 10 hosts sending packets at the same moment if the probability for each host to send packet is 0.1?

...

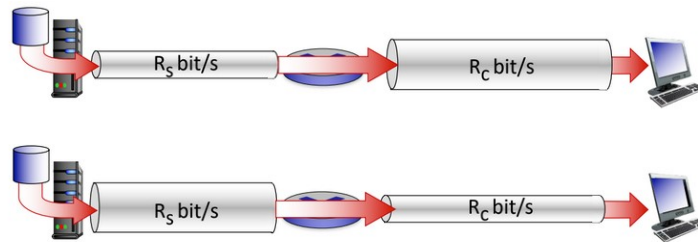
Connection performance

bandwidth: With bandwidth we refers to the width of the wave signal in hertz, the more is wide this the more is the information that a signal can carry, and "bit-rate" the quantity of bit that a connection can exchange in a slice of time.

| The first is referred to a transmission channel, the second to a link.

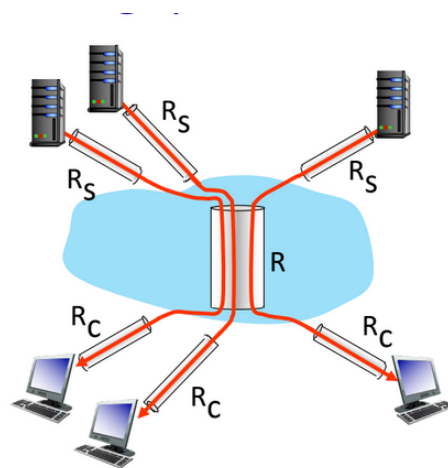
throughput: is the quantity of bits that pass through the link in a point of the network, its limit is the bit-rate.

If a packet has to pass through different bit-rate links there is a bottle-neck, $\max \text{throughput}(\text{fast link}) = \max \text{throughput}(\text{slow link})$.



Max bit-rate of the network = worst bit-rate of the network.

- Is given the following situation:



Max bit rate = $\min(R_C, R_S, R/\text{number of connetions})$

Delays

In a communication we have 4 different delays.

- Dproc = time to verify that there aren't problems on a node (transcurable).
- Dqueue = amount of time that a package stays in the router queue.
- Dtrans = amount of time to transfer data on the link.

- D_{drop} = amount of time that a package spends to cross the link from a node to an other node.

- D_{trans} , is measured in L/R (L = length of a package in bit, R = bit-rate)

- D_{prop} , is measured in k/v (k = is the physical length of a link, v the physical speed of a connection)

D_{queue} , is measured in $L*a/R$ and it is limited by the time unity. (L is the length in bit of a package, a is the tax of arrival of new package, R is Bit rate on the link)

if $(L*a/R) \geq 1$ the D_{queue} tends to infinite

else it exponentially encreases.

- Bit-rate $\times D_{\text{prop}}$ = amount of bits that a link can contain.

Exercises:

[esercizi su tempi di trasferimento.xopp](#)

[esercizi su tempi di trasferimento..pdf](#)

[esercizi queuing.xopp](#)

[esercizi queuing.pdf](#)

Protocols

A protocol is a set of rules that controls the type of package exchanged or messages exchanged between peer entities (two entities is peer when has the same level on the protocol hierarchy).

A service is a set of primitives.

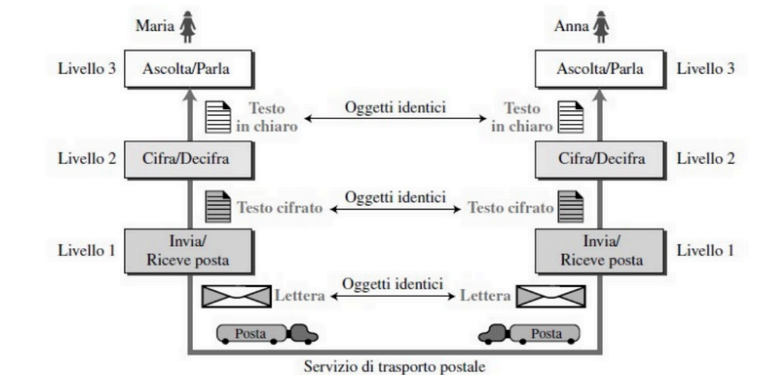
Protocol hierarchy

A network communication is a very complex system, protocols bornt to simplify it. Communication is divided in layer, each layer has a specific protocol, has specific tasks and offers services for the layers above, only peer communicate using a protocol. (each layer uses the services of the layers below and offers services to the layers above).

The scope of the layered structure is to offer services to layers above other layers hiding the implementation details of the services.

The layering of the communication and the introduction of protocols, simplifies the model and make more manutenable the system, the only malus is the decline in performance due the redondance of the operations (incapsulation and decapsulation to exchange packages)

Example:



layers are bidirectional, for example the layer that encrypt data has to be able to decrypt data.

Network is divided in 5 macro-levels:

Application
Transport
Network
Link
Physical

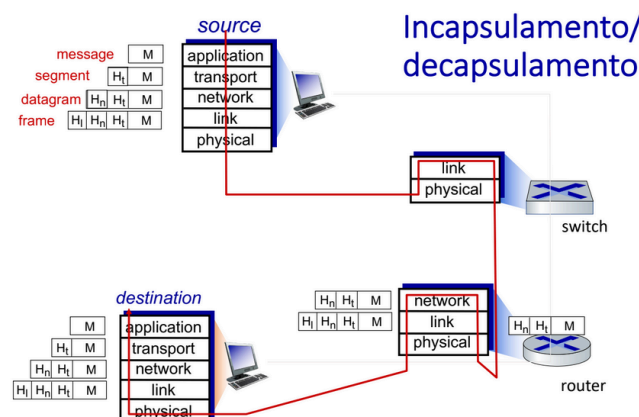
- Application level. is where are applications that use Internet services, protocols at this level are HTTPS, SMTP, FTP, DNS... in this level, packages are called messages.
- Transport level. is the level where host's messages is converted to be read by a server, important protocols at this level are TCP, UDP...in this level packages are called segments
- Network level. network level route the segments from a sender to a recipient. An important protocol at this level is IP, in this level packages are called datagrams.
- Link level. It deals the transimtion of datagrams from a node to a node, important protocol at this level is Ethernet, Wi-Fi... in this level packages are calles frames.
- Physical level. It deals how bits can be physically stored and physically transferred.

Incapsulation/Decapsulation

If you want to send a package it has to be incapsulated.

- From application level to transport level is added to the package a "transport header"
- From transport level to network level is added to the message a "network header"
- From network level to link level is added to the datagram a "link header"
- From link level to physical level is added to the frame a "physical header"

When the frame arrives to the recipient it has to decapsulate this frame to load the data packet.

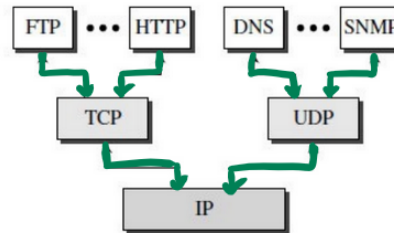


In this example when the frame arrives to the switch, the switch doesn't fully decapsulate the frame, It needs only some levels

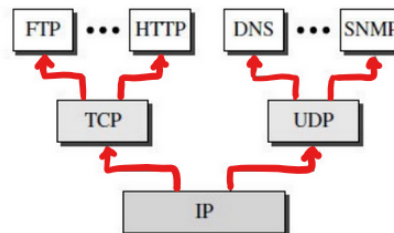
Multiplexing/de-Multiplexing

In each level there are different protocols, for this reason is fundamental the multiplexing on the source and the demultiplexing on the destination.

- Multiplexing: more protocols of a level are encapsulated with the same protocol in the level below.



- de-Multiplexing: a protocol in a level is decapsulated into more protocols.



To do multiplexing and demultiplexing operations each data packet must have in the headers a field that specifies the used protocol.

Introduction to Cyber-security

there are a lot of threat in Internet.

- malware: A software that access to pcs thanks to emails or executable files, it is autoreplicant because it can send emails.
- spyware: A software that spy your visited web sites. the user that is infected can take part of a botnet, a network of pcs with the same spyware that can be used for bad uses by an hacker.
- DOS attack: (Denial of Services) is an attack where an hacker sends you a lot of package to create traffic on the link to make unavailable a service
- Packet sniffing: Is an attack where an hacker can see and de-encrypt the packages that you send or recieve.
- IP spoofing: An hacker uses a fake ip (an ip that you know) to steal your information.

There are a lot of tecnics that we can use to defend our data.

- autentification (reveal the real identity of someone).
- confidentiality (encrypt messages).
- integrity check (associate digital firms to a packet to avoid to load compromitted packets).
- firewall, a program in the access point that filters packages and stops DOS attacks.
- access restriction, VPNs protected by passwords...

Osi model

in this model we have two more level, presentation level and session level.

- presentation level allows to an application to interpreates the meaning of data (cryptografy)
- session level deals the sincronization.

Application
presentation
session
Transport
Network
Link
Physical

Application Level

CLIENT-SERVER PARADIGM

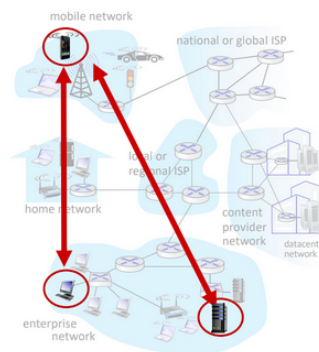
Clients and servers in this paradigm is two type of entities.

- SERVER

A server has a fixed IP and it cannot change and it is always on.

- CLIENT

A client in this paradigm communicates with servers and it cannot communicate with other client, it can have dynamic IP and it isn't always on.



PEER-TO-PEER PARADIGM

A client is also a server and a server is also a client, two client in this paradigm can directly communicate each other, they have dynamic IP.

There is only one problem, a BIG problem, this type of connections is very difficult to manage.



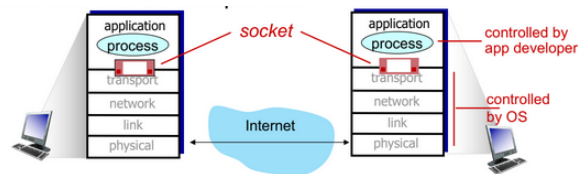
Communication between processes

Processes in a client are different to processes in a Server, this is not true on the P-2-P paradigm, two processes on the same host communicates each other with the inter-process(a system defined by OS)

while to communicate between two different hosts they exchange messages.

Socket :

A process that wants to exchange messages (receives/sends) has to open a socket, a socket is a sort of portal, the sender process sends messages through its socket while the recipient process takes that messages from its socket.



To route a message the recipient must have a unique IP (32 bits) and the door number because more processes can run on the same host.

example: the door number for an http server is 80

the door number for a mail server is 25...

An application level protocol defines:

- message type, for example: request, answer, comment...
- syntax of the message, the fields in a messages
- semantic of a message, the meaning of the information in the fields.
- rules: when send messages?...

some protocols like HTTP are free, others are private like the protocol of [Skype \(MICROSOFT\)](#).

An application protocol requires some services from the level below.

- Data integrity: no packet loss
- temporal warranty: low ping
- throughput: good connection speed
- security

application	data loss	throughput	time sensitive?
file transfer/download	no loss	elastic	no
e-mail	no loss	elastic	no
Web documents	no loss	elastic	no
real-time audio/video	loss-tolerant	audio: 5Kbps-1Mbps video: 10Kbps-5Mbps	yes, 10's ms
streaming audio/video	loss-tolerant	same as above	yes, few s
interactive games	loss-tolerant	Kbps+	yes, 10's ms
text messaging	no loss	elastic	yes and no

There are two principal transport protocols

- **TCP**

Services:

1. **Reliable transport**: the protocol checks if the sent message is received by the recipient
2. **Flow control**: the protocol stops the sender if it overloads the network
3. **Overload control**: the protocol limits the sender if the network is in overload
4. It requires a configuration between client and server (**HANDSHAKE**)
5. It doesn't have systems for minimal throughput, security and timing (low ping).

- **UDP**

Services:

no services

but

it is faster

and

it doesn't requires a configuration between client and services

Functionality like Flow check, Overload control... are implemented in software level with libraries (for example TLS)

Security

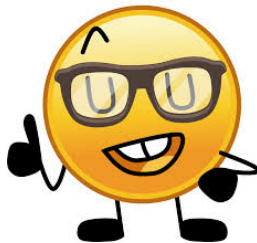
TCP and UDP doesn't have security, this is very dangerous for users, for example they don't encrypt passwords!, there is a solution:

- TLS

TLS is a software library that encrypt/decrypt data, this library is used in the application level.

HTTP

▼ Hyper-text transfer protocol 🤓



A web page includes web objects for each of these there is an univoc url.

url: uniform resource locator 🤓

HTTP

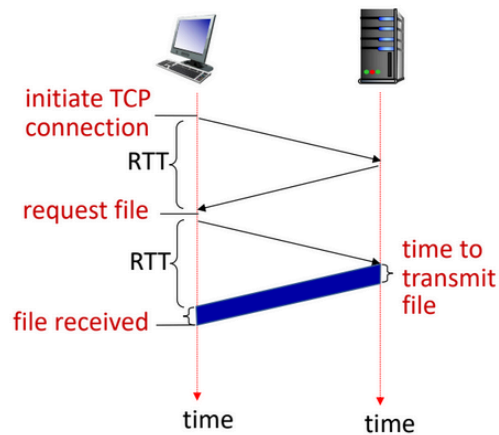
- Is the most famous and used application level protocol
- It uses TCP as transport protocol
- Is a client-server protocol where
 - Client: browser that requests, recieve web objects (jpeg, html page, png, mp3 file)
 - Server: the server sends web objects as answer for requests.
- http is stateless, it don't preserve data of clients

HTTP 1

In http1 we have http non-persistent:

for each object in the request, a TCP conection is opened, when the object is received by the client, the TCP connection is closed, when a request has more object than one, the http1 protocol open a TCP connection for each object in the request. It is very unefficient because for each object in the request we have 2 RRT delay that is the delay to send a small packet from client to server and from server to client.

RTT : round trip time 🤓

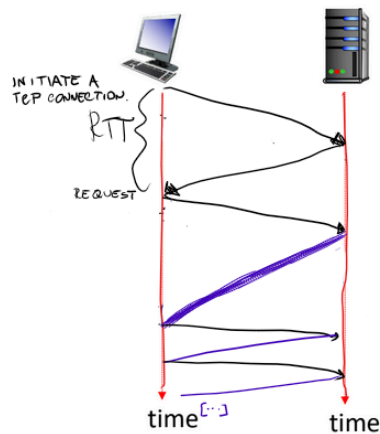


When we request a web object with http1 protocol the delay will be $2RTT + \text{delay to transmit file}$.

1 RTT to initialize a TCP connection, 1 RTT to send http request.

HTTP 1.1

In this protocol the TCP connection stays opened, in this modality for each object in a request we have only one RTT for object



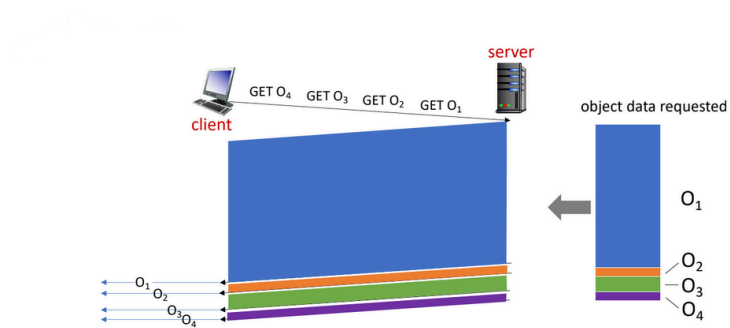
HTTP 1 = 2 RTT for each object.

HTTP 1.1 = 1RTT for each object.

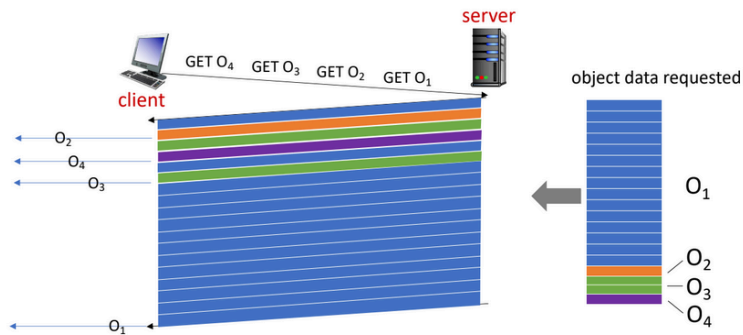
The time for a request is approximately halved.

HTTP 2

In http1.1 we have a big problem, the object are loaded in randomically order, if in a request there are object of very different size and the biggest is scheduled for first, the smaller have to attend a lot of time and, if there is a packet-loss the big object need to restart the transfer. (A LOT OF WASTED TIME)



With http 2 protocol, the objects are divided in frames and and these frames are interlaced.



However starting a data load again due to packet-loss remained as problem in this version

HTTP 3

In this version TCP is abandoned as transport protocol, in fact HTTP3 uses UDP and is added more security and error check... (application level)

HTTP Request message

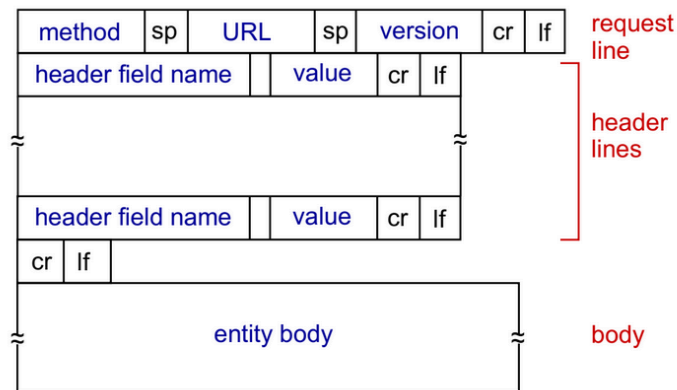
There is two types of messages in HTTP protocol, these is request and answer.

- Request message
 - it is written in ASCII format
 - The request message consists of three sections: the request line (which includes the method, URL, and HTTP version), the header fields (which provide additional information about the request), and the optional message body (which can contain data or documents to be sent to the server).
 - famous headers:
 - User-agent: Indicates the utilized client program (firefox, chrome...)
 - Accept: Indicate the output data type accepted
 - Accept-charset: Family of chars that the client can leverages
 - Accept-encoding: encoding scheme utilized by the client
 - Accept-language: Favorite language of the client (English, Italian...)
 - Authorization
 - Host: name of the host and port number
 - Date: Date
 - Upgrade: Specifica il protocollo di comunicazione preferito
 - Cookie: communicate a cookie to the server
 - If-Modified-Since(date): send document only if is more recent of the specified date

Example:

REQUEST LINE → GET /index.html HTTP/1.1\r\n
HEADER → Host: www-net.cs.umass.edu\r\n
 User-Agent: Firefox/3.6.10\r\n
 Accept: text/html,application/xhtml+xml\r\n
 Accept-Language: en-us,en;q=0.5\r\n
 Accept-Encoding: gzip,deflate\r\n
 Accept-Charset: ISO-8859-1,utf-8;q=0.7\r\n
 Keep-Alive: 115\r\n
 Connection: keep-alive\r\n
END → \r\n
INDICATED WITH TWO NEW LINES.

General format of the request:



Methods in http request:

GET with parameters:

To send data to server, this is insert after the char '?'

PUT

to upload a new web object on a server, if the url exist yet the new object will substitutes the older (very dangerous, now this operation is limited)

POST

to request data to server

HEAD

to request the header information of the object requested.

- **Answer message**
 - it is also in ASCII
 - consists in three different parts, state, header and data
 - the header provides more information about the result.
 - **Header:**
 - Date: Date
 - Upgrade: specifies the favorite protocol
 - Server: Indicates the server program
 - Set-Cookie: server requires to the client to store a cookie
 - Content-Encoding: Specifies the codification type
 - Content-Language: Specifies the language of the file
 - Content-Length: Indicates the size of the file
 - Content-Type: Indicates the type of the file
 - Location: ask to client to send request to another site

Last-Modified: Indicates the date and the time of the last modify on the document.

STATE → HTTP/1.1 200 OK\r\n
Date: Sun, 26 Sep 2010 20:09:20 GMT\r\n
Server: Apache/2.0.52 (CentOS)\r\n
Last-Modified: Tue, 30 Oct 2007 17:00:02 GMT\r\n
HEADER → ETag: "17dc6-a5c-bf716880"\r\n
Accept-Ranges: bytes\r\n
Content-Length: 2652\r\n
Keep-Alive: timeout=10, max=100\r\n
Connection: Keep-Alive\r\n
Content-Type: text/html; charset=ISO-8859-1\r\n
DATA → \r\n
data data data data data ...
EX: HTML, JAVA APPLET ...

- State has a code, each code indicates some about the request.

most famous code:

```
200: ok
204: no content present (there is nothing to do)
301: moved permanently
400: bad request (the sended message is not understud by the server)
404: not found (the url doesn't exist on the server)
503: try again later
505: http version not supported (the server not support http client version)
```

in general:

Code	Meaning
1xx	information
2xx	success
3xx	redirection
4xx	client error
5xx	server error

Mantein the client state on a server

We said that the http protocol is state-less it don't presererve the previous request or information about client.

One of the problem of a protocol with state is the partially completed operation, in certain case the server has to restore data at the state that they before the operation.

Therefor, in certain case the manteinance of informations is fundamental (amazon kart, web e-mail, authorization for a web-site), data can be stored using **cookies**.

| http rest state-less, http requests are indipendent each other.

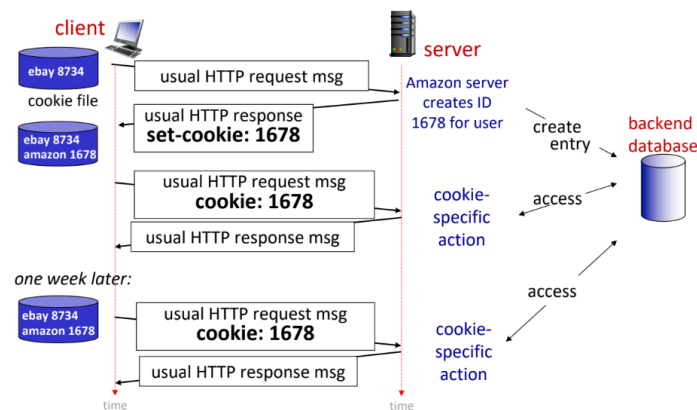
cookies:

Cookies is files stored on clients that servers use to identifies who is the client and retores its information.

To implement cookies on a site, the site must have 4 components:

1. http header message that indicates that the cookies are set for the client
2. http header message that request for cookies
3. a database and a back-end section
4. cookie file stored on the client data usable for the browser

example:



Cookies has variable life-time. when a client closed a season the server set the cookie with a max age, Max-Age(in seconds) (max age can be 0 seconds) after the time expressed in Max-Age the client deletes the cookie.

Cookie is also a dangerous system, they can learn a lot about a client, violates his privacy or suggest indesidered advertaising.

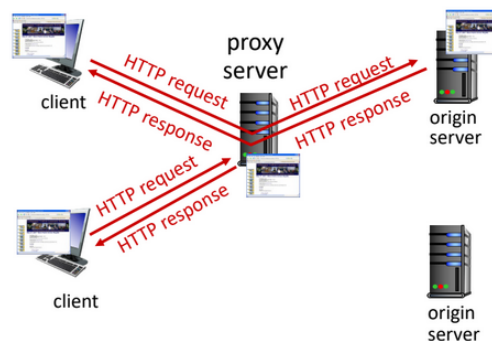
Proxy servers (web cache)

proxy servers are server used as web cache, the goal of a proxy server is to involve less as possible the main server.

An user try to take data from a proxy server, if data are stored on the proxy server, the proxy server sends data to the user, else the proxy searches these data on the main server, the proxy copies these data on its hard drives and after this operation, it send data to the client.

It is very useful because reduces the traffic on the main line, reduces the response because it is more close to the client and make more responsive important and "popular data".

for example in a university the calendar will be very responsive because a lot of people search for calendar and it is in proxy.



Elettronic-Mail

- user-agent (UA)

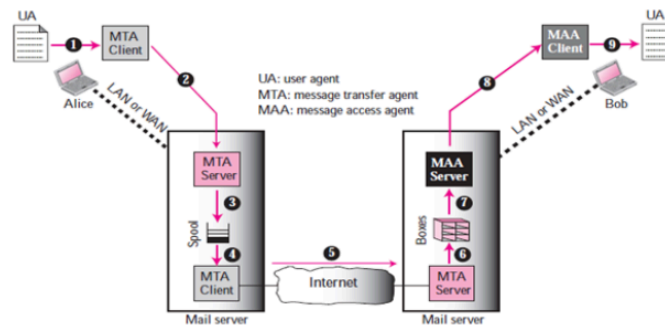
the user-agent or (mail reader) is a software activated by the user or to a timer, with this, users can read and write mails to send to other users.

example of UA are Outlook, Eudora, Gmail...

The messages in exit are stored on a MTA server.

- mail Transfer agent (MTA) is a process active on a mail server that transfer data from an UA or from an other MTA

- mail access agent (MAA), is a process active on a mail server that allows you to read emails.



PROTOCOLS FOR eMAIL.

- SMTP

Simple mail transfer Protocol.

This protocol uses a directed transferring, it transfers mail from its mail server to the mail server of the recipient.

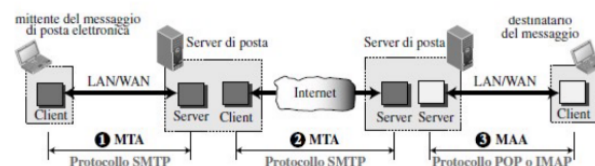
the client send messages in ASCII codification (Header and body) and the server answer to the user in ASCII codification

how it performs?

1. This protocol uses TCP as transfer protocol, it open a connection with the server of the recipient (handshaking) on the socket port 25, If there are problem during the connection the client will retry after few seconds.
2. In a second phase client and server do a second handshaking and the client provide the email address of the recipient, messages are sent
3. the TCP connection is closed.

example:

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250 Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
```



protocol to read message is not SMTP but POP3 or IMAP.
to read about POP3 and IMAP

Click in the link below

<https://www.notion.so/Computer-networks-d06f3126c5c24917b540a99c5c464524?pvs=4#27d518a899894b91812a1fb6d5b09fcb>

There is a standard that a mail message has to following: RFC 822

- RFC 822

A mail message in RFC 822 standard has an header and a body.

The header has the following fields:

To: recipient mail address

From: your mail address

CC: people to advise of the exchange of the message

BCC: people to advise of the exchange of the message but the recipient cannot see these

Subject: the title of the message, the argument

Sender: the name of the sender

The body of the message has to be in ASCII.

Other standards are RFC 2045 and RFC 2046, They define the MIME standard (multipurpose internet mail extension) message in this standard are encrypted when the message is send and decrypted by the recipient. in The header must be a line "Type" that indicates the type of files in the message.

POP3 AND IMAP.

- POP3

post office protocol 3

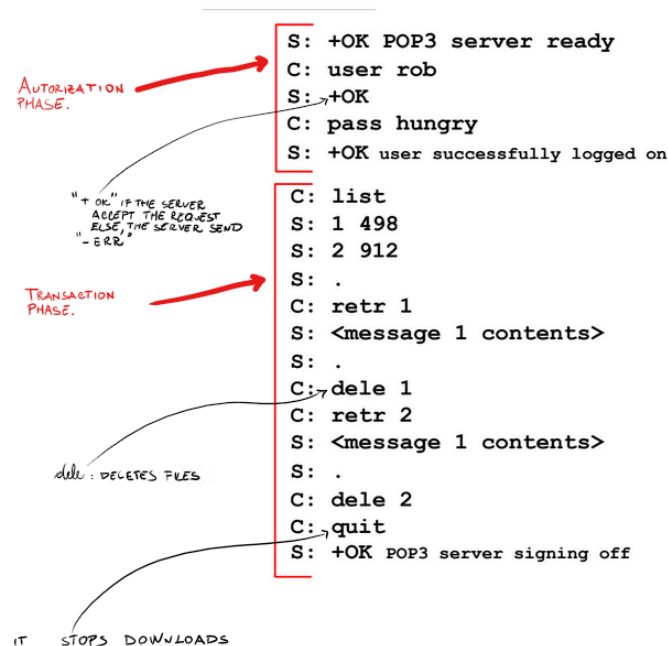
Is a simple protocol used to download mail messages.

It is a state-less protocol, uses TCP protocol and the 110 door of the socket.

How it performs?

1. Connection: The client open a TCP connection with the mail server. (Handshaking)
2. Authorization: The client (UA) send user name and password to be identified
3. Transmission: The client (UA) downloads the messages from the mail server
4. Update: after the "QUIT" message all the downloaded files are removed from the server.

example:



POP3 is a very limited protocol, you cannot create portable folders or mantain files on the server.

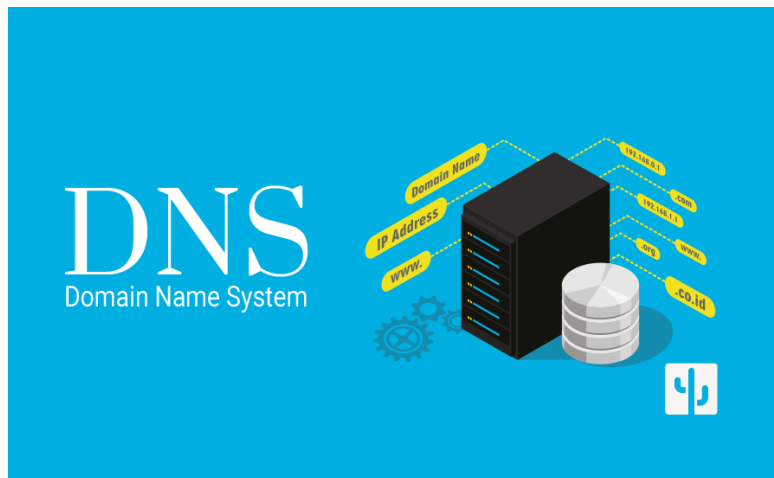
- IMAP

In IMAP protocol the problems of POP 3 are solved because files are stored on the server (not on client). it uses a stateful protocol (a protocol with a state), the client can create folders with name.

HTTP

Some mail server uses http protocol, SMTP is required to communication between mail servers.

DNS



Person has a lot of univocal identifier

example: name, surname, matricola..

Hosts have univocal identifiers that are:

- IP address (32 bits)
- name (ex. cs.umass.edu..)

how we can map the ip address to a the name of an host?

The answer is: "DNS"

DNS is network of database implemented as a hierarchy of "name servers" that uses a application level protocol to "solve" names.

This is a core function of internet.

Functions of DNS:

- Aliasing

This function allows site to has a more simple name

example:

relay1.west-coast.enterprise.com (canonical)

can has these alias:

www.enterprise.com or enterprise.com (alias)

DNS can be used to translates canonical name to an alias name or to translate an alias to a canonical from an alias or to have the IP of the host.

- Load distribution

sites with a lot of traffic are replicated on a lot of servers, they have different ip addresses and run on different hosts, DNS can associate the same name to more ip addresses. When an host call the DNS to solve a name the DNS return one of the IP address in the list of ip addresses with (associated to) the same name, This strategy reduces the traffic on a single host.

This Strategy solves a lot of problems (PING).

DNS construction

A DNS is big a decentralized database, it has billion of records (simple records)

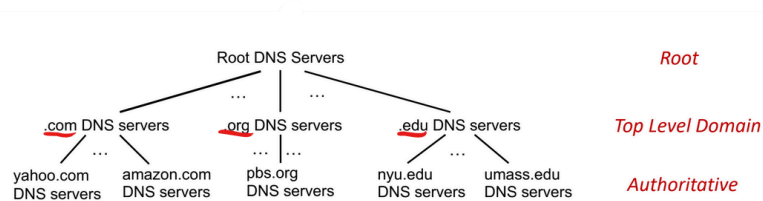
A DNS has a lot of requests (millions of request every day)

It is a core function of internet, DNSs is very secure and reliable

there is a strong security system, the DNSSEC that release message authenticity and integrity.

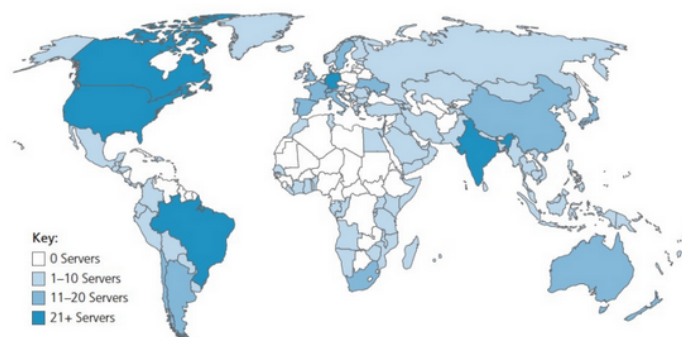
Hierarchy:

A single DNS server cannot contain all the ip addresses in the world (2^{32} addresses), these addresses are distributed on more DNS servers.



There is a main server (ROOT DNS server) replicated a lot of times,

these are connected to various Top level Domain servers (replicated) and these are connected to smaller DNS called authoritative.



there are 13 principal root name server, (13 logical root server) each server is replicated a lot of times.

If a client want the ip address for www.pornhub.com

- Using UDP the client calls the root server to found .com (TLD) DNS.
- The client contact .com (TLD) server to obtain the IP of the pornub's authoritative server
- The client using UDP protocol calls the authoritative server of pornhub.com.
- The client calls the authoritative server to obtain the ip for www.pornhub.com.

- **Top Level Domain (TLD)**

Example: .com, .org, .net, .edu, .aero.. or .uk, .fr, .it, .jp..

- **Authoritative DNS server**

Each organizationan has a server, it is kept by the organization or by a releaser.

Often each organization has two server (PRIMARY and SECONDARY server).

- **Local DNS server**

This type of server aren't in the hierarchy.

Each ISP has a local DNS server called default name server.

It has the function of a proxy (used as cache). Records in the cache will be refreshed after a TTL (Time to live) or manually with a command. If the ip of a server change it will not update automatically on the cache server.

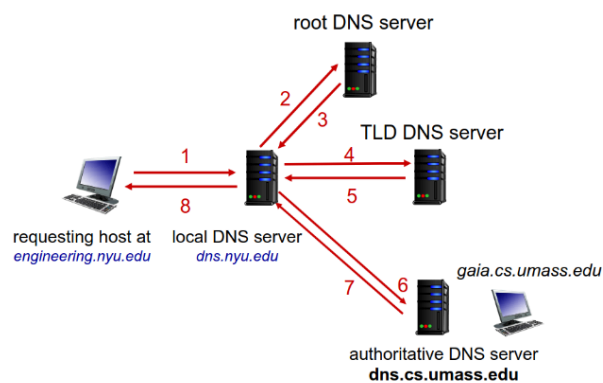
COMMON DOMAIN LABELS

LABEL	DESCRIPTION
aero	flying company
biz	similar to com
com	commercial company
coop	cooperation
edu	educative institution
gov	governative institution
info	releaser of informative services
int	international organization
mil	military organization
museum	museums
name	person name
net	network organization
org	organization with no lucre scope
pro	professional organization
it	italy
fr	france
uk	uk
jp	japan

Name resolution DNS:

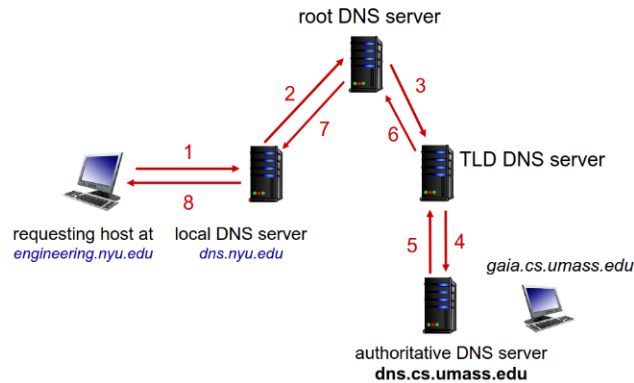
Iterative query

In this schema when a server is contacted by a client it return the next server to contact to the client until the client contacts the solver server.



Recursive query

In this schema the resolution of a name is committed to the contacted server recursively



In this example the requesting host interrogates its local DNS, it doesn't have the ip for the name released by the requesting host, it search it on the DNS root, the root DNS server search the ip on the TDL .edu server, the TDL .net server found the authoritative DNS server.

Each map address-name is insert in a resource record (RR) that has the following structure (name, value, type, ttl)

- type = A : "name" field contains the name of an internal host in a domain (hostname) and the value field contain the IP address of this host.

example: name: raly1.bar.foo.com, value = 45.37.93.126

- type = NS : "name" field contains the name of a domain and the value field contains the host-name of the authoritative server.

example: name: foo.com, value = dns.foo.com

- type = CNAME : "name" field contains an alias while the "value" field contains the canonical name.

example: name: www.ibm.com, value = servereast.backup2.ibm.com

- type = MX : used for mail server, the "name" field contains the name (server1diCarlo) of the server while the "value" field contains the host-name of the server.

DNS records

DNS: distributed database storing resource records (RR)

RR format: (name, value, type, ttl)

type=A

- name: hostname (relay1.bar.foo.com)
- value: IP address (45.37.93.126)

type=NS

- name: domain (foo.com)
- value: hostname of authoritative name server for this domain (dns.foo.com)

type=CNAME

- name: è l'alias nel vero nome (canonico)
- www.ibm.com in realtà è servereast.backup2.ibm.com
- value: canonical name

type=MX

- value is name of mailserver associated with name

Example:

In the Sapienza server (an authoritative server), there is an A type record:

(corsi.di.uniroma1.it, 131.111.45.68, A)

In the .it DNS server (a non-authoritative server), there is an A type record and an NS record:

(uniroma1.it, dns.uniroma1.it, NS)

(dns.uniroma1.it, 128.119.40.111, A)

According to this schema, a client contacts the root server for corsi.di.uniroma1.it. It returns the .it DNS, which contains the name and IP address of the DNS for uniroma1.it (dns.uniroma1.it).

The authoritative server contains the IP for corsi.di.uniroma1.it (131.111.45.68).

DNS PROTOCOL

DNS protocol is an application level protocol, it utilizes UDP as transport protocol because it faster (is not required any handshakes).

DNS is state-less because don't need to save state.

- **MESSAGES**

Answers and requests has the same format, an header of 32 bits

16 bits for the identification

16 bits for flags

and vary fields.

- The 'identification' field contains information about the applicant.
- The 'flags' field contains various indicators:
 - A flag that indicates whether the message is an answer or a question.
 - A flag that indicates whether recursive resolution is allowed or preferred.
 - A flag that indicates if the answer is from an authoritative server.
- The 'questions' field of variable size contains information for a question.
- The 'answers' field of variable size contains information for an answer (RR).
- The 'authority' field contains the authoritative RR to send as an answer.
- An extra field is provided for additional information.

